

Important Facts of Photoelectric Effect

1. Why does increasing light intensity increase current but NOT the maximum kinetic energy?

What classical physics expected

According to **classical wave theory of light**:

- Light is an electromagnetic wave.
- Energy of a wave is spread continuously over the wavefront.
- **Intensity** of light = energy delivered per second per unit area.

So classical physics predicted:

If you increase intensity, more energy should hit each electron.

Therefore:

- Electrons should absorb more energy
- Electrons should come out with **larger kinetic energy**
- Even low-frequency light should eventually eject electrons if intense enough

But experiments showed:

- Increasing intensity → **more electrons emitted**
- But **maximum kinetic energy stays the same**

This was a huge contradiction.

Einstein's explanation (photon picture)

Einstein said: Light does **not** deliver energy continuously.

Instead: Light comes in packets (quanta), later called **photons**

Each photon has energy:

$$E=hf$$

where

h = Planck's constant

f = frequency of light

Key idea:

A single electron absorbs one photon

Not many little pieces continuously.

So:

Case 1: Increase intensity at same frequency

Suppose red light frequency is fixed.

Photon energy is:

$$E = hf$$

Since frequency is unchanged:

- Energy of each photon is unchanged

But intensity means:

- More photons arriving per second

Thus:

Incident	Results
More photons	More electrons emitted
Same energy per photon	Same KE of each electron

Hence:

- **Current increases**
- **Maximum kinetic energy does NOT increase**

Mathematical explanation

Einstein's equation:

$$hf = \phi + K_{\max}$$

where

- hf = photon energy
- ϕ = work function (minimum energy needed)
- $K_{\max}$ = maximum kinetic energy of electrons

So:

$$K_{\max} = hf - \phi$$

Notice:

There is NO intensity term here!

Kinetic energy depends only on:

- frequency f
- work function ϕ

NOT on intensity.

Why current increases

Photocurrent depends on number of electrons emitted per second.

If intensity increases:

- number of photons increases
- more electrons are struck
- more electrons escape

So:

Intensity $\propto$ number of photons

Thus:

Intensity controls current, frequency controls energy.

Simple analogy

Imagine photons as **bullets** and electrons as **balls on a table**.

Frequency = bullet power

Higher frequency $\rightarrow$ more energetic bullet.

Intensity = number of bullets

More bullets $\rightarrow$ hit more balls.

If you fire many weak bullets

- More balls may be hit
- But each bullet has same energy
- No ball gets extra speed beyond what one bullet can provide

That is why:

- More intensity $\rightarrow$ more electrons
- But same KE

From Planck's idea to Einstein's Nobel-Prize explanation

This is one of the greatest developments in physics.

Let us go step by step.

STEP A: Problem of blackbody radiation

In the late 1800s physicists studied **blackbody radiation**.

A hot object emits radiation of many frequencies.

Experimental graph:

- low frequency $\rightarrow$ low energy
- medium frequency $\rightarrow$ peak
- high frequency $\rightarrow$ falls

Classical theory predicted something terrible:

At high frequency:

$$u(f) \rightarrow \infty$$

Infinite energy!

This disaster was called: Ultraviolet Catastrophe

Why classical theory failed

Classical equipartition theorem said:

Every mode of oscillation gets:

$$kT$$

So infinitely many high-frequency modes would contain energy.

Thus:

$$u(f) \propto f^2$$

which diverges at high frequency.

Experiments did NOT agree.

STEP B: Planck's bold assumption (1900)

Max Planck tried to fix this.

He made a radical assumption:

Atoms in cavity walls do NOT emit energy continuously.

They emit in packets: $E = nhf$

Where, $n=1,2,3,\dots$, h = Planck's constant, f = oscillator frequency

Thus energies allowed:

hf , $2hf$, $3hf$...not any arbitrary value.

This was the birth of **energy quantization**.

What Planck actually thought

Important subtle point:

Planck did **not** say light itself is particles.

He said:

Oscillators in matter exchange energy in discrete amounts.

In his mind:

- Radiation could still be wave-like
- Quantization was a mathematical trick to fit data

Planck himself did not fully believe energy packets were physically real.

Planck's formula matched experiments perfectly.

Significance

Planck introduced:

$$E = hf$$

But he applied it to:

oscillators in matter

not to light itself.

STEP C: Photoelectric effect creates another crisis

Experiments found:

Observation 1

No electrons emitted below threshold frequency.

Even intense light fails.

Observation 2

Above threshold frequency:

electrons emitted instantly.

No time delay.

Observation 3

Increasing intensity:

- increases current
 - not KE
-

Observation 4

KE depends linearly on frequency

$$K_{\max} \propto f$$

Classical theory could not explain any of these.

STEP D: Einstein's revolutionary leap (1905)

Albert Einstein did something much more radical.

He said:

Planck's quantum idea is not just about matter.

Light itself behaves as if it consists of localized energy packets.

He called them:

light quanta (later photons)

Each has energy:

$$E=hf$$

This was revolutionary.

Einstein's connection to Planck

Planck said:

Matter emits/absorbs energy in quanta.

Einstein said:

The radiation field itself is quantized.

That was a giant conceptual leap.

STEP E: Einstein's photoelectric equation

When a photon strikes an electron:

$$hf = \phi + K_{\max}$$

So:

$$K_{\max} = hf - \phi$$

This explained ALL experiments

1. Threshold frequency/Cut off frequency

Need minimum energy:

$$hf_0 = \phi$$

Thus:

$$f_0 = \phi$$

If: $f < f_0$ then: $hf < \phi$

No electron escapes.

Even huge intensity cannot help.

2. Instant emission

Photon gives energy in **one collision**

No waiting needed.

Electron gets all energy at once.

3. Intensity increases current only

Same frequency means same photon energy

More photons means more electrons

4. KE depends on frequency

$$K_{\max} = hf - \phi$$

So, $K_{\max}$ vs f graph is nothing but a Straight line.

Slope: h

Intercept: $-\phi$

Exactly what experiment showed.

Why this was so revolutionary

Classical wave theory said:

Light = wave only.

Einstein said:

Light behaves as if it has **particle-like quanta**.

This introduced:

wave-particle duality

which later became central to quantum mechanics.

Why Einstein got the Nobel Prize

Many people think Einstein got Nobel Prize for relativity.

He did **NOT**.

He received the **1921 Nobel Prize in Physics** for:

“his services to theoretical physics, and especially for his discovery of the law of the photoelectric effect”

Nobel Prize in Physics 1921

Because:

- He gave the first correct explanation
- He showed Planck's quantum had real physical meaning
- This helped create quantum mechanics

Summary: Planck vs Einstein

Planck

Matter oscillators quantized

$$E = nhf$$

Used for blackbody radiation

Semi-mathematical assumption

Did not claim photons

Einstein

Light itself quantized

$$\text{photon energy } E = hf$$

Used for photoelectric effect

Physical interpretation

Introduced light quanta

One sentence summary

Planck quantized the exchange of energy in matter; Einstein extended that idea and said light itself carries energy in discrete packets (photons), which perfectly explained the photoelectric effect and launched quantum physics.